Evaluation Traps in EEG Disease Classification: Identity Leakage, Lucky Folds, and Objective Mismatch Replicated Across AD, FTD, MDD, and SCZ

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Abstract

Small-cohort clinical classification studies in neuroscience and healthcare routinely report accuracy figures that cannot be reproduced under rigorous evaluation. EEG disease detection is an exemplar of this broader problem, and what happens here is likely happening across neuroimaging and clinical machine learning more broadly. Across five resting-state cohorts spanning Alzheimer's disease, frontotemporal dementia, major depressive disorder, and schizophrenia, we identify three structural evaluation traps. **Trap 1 (identity leakage):** when epochs (fixed-length segments cut from a continuous EEG recording) from the same subject appear in both train and test sets, accuracy is inflated. **Trap 2 (lucky folds):** even with subject-disjoint splits, results can still vary strongly depending on which subjects are held out, with some folds reaching perfect accuracy. **Trap 3 (objective mismatch):** epoch accuracy and subject accuracy can rank models differently. All three traps replicate across every cohort regardless of biomarker strength, confirming they are structural properties of evaluation design. We propose a four-point reporting standard and release all prepared data and code in an accessible and extensible format.

1 Introduction

EEG disease classification studies routinely report accuracy figures that do not replicate under rigorous evaluation [4, 3]. Small sample sizes in neuroscience are known to inflate effect estimates and reduce reproducibility [6, 5]. EEG is an ideal test case for this audit because subject pools are small, recordings are long and generate hundreds of epochs per subject. The consequence is direct. Without reproducible results, there is no reliable basis for knowing which models generalize to new patients, which papers to trust, or which methods are ready for clinical use. Miltiadous et al. [2] showed that validation choice alone inflates accuracy on AD/FTD data. A recent large-scale benchmark [21] found accuracy clustering at 0.53–0.84 under strict subject-disjoint evaluation, which is far below the 90–99% figures common in many studies. No prior work has quantified the sources of this gap across multiple diseases in a single auditable framework.

To address this, we identify and quantify three structural evaluation traps across five resting-state EEG cohorts (AD, FTD, MDD EC, MDD EO, and SCZ) using identical pipelines and 13,200 independent fit-transform-predict cycles. **Trap 1 (identity leakage):** when epochs (fixed-length segments cut from a continuous EEG recording) from the same subject appear in both train and test sets, accuracy is inflated. **Trap 2 (lucky folds):** even with subject-disjoint splits, results vary strongly depending on which subjects are held out, with some folds reaching perfect accuracy. **Trap 3 (objective mismatch):** epoch accuracy and subject accuracy can rank models differently. All three traps replicate across every cohort regardless of biomarker strength. For Trap 3, what is consistent is not the direction of the

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gap but the unreliability of using one accuracy as a proxy for the other. We release all split manifests, pipeline configurations, and source code at github.com/anon333science2026-afk/eeg-evaluation-traps.

2 Datasets and Preprocessing

Datasets. Three independent datasets provide five cohorts (Table 1). The AD and FTD cohorts come from OpenNeuro ds004504 and ds004904 [1, 15]. The MDD cohort comes from Mumtaz et al. (2017) [22], analysed separately under eyes-closed and eyes-open conditions. The SCZ cohort comes from Olejarczyk & Jernajczyk (2017) [23]. Dataset curation, recording parameters, and BIDS packaging details are in Appendix A.

Table 1: Five cohorts used in this study. Ep. chance and Subj. chance are majority-class baselines at epoch-level and subject-level respectively.

Dataset	Disease	Control	Total epochs	Mean $\pm$ SD	Ep. chance	Subj. chance	Source
AD (eyes-closed)	36	29	34,044	524 ± 86	0.547	0.554	OpenNeuro ds004504
FTD (eyes-closed)	23	29	25,881	498 ± 95	0.597	0.558	OpenNeuro ds004904
MDD (eyes-closed)	29	27	10,978	196 ± 16	0.503	0.518	Mumtaz et al. (2017)
MDD (eyes-open)	31	28	11,539	196 ± 16	0.520	0.525	Mumtaz et al. (2017)
SCZ (eyes-closed)	14	14	18,535	662 ± 108	0.535	0.500	Olejarczyk & Jernajczyk (2017)

Preprocessing. All signals are preprocessed in Python using MNE-Python [16]. A 0.5 Hz high-pass filter removes slow drift; a 30 Hz low-pass filter defines the analysis band. Because the analysis band is capped at 30 Hz, mains interference at 50/60 Hz is excluded by design. No notch filter is applied.

Epoching. EEG recordings are segmented into 3-second epochs with 50% overlap (stride = 1.5 s).

Artifact rejection. Epoch rejection is automated and deterministic. Epochs exceeding 800 μV amplitude or falling below 0.3 μV are rejected. Dataset annotations are respected where available. No manual inspection steps are used.

Feature extraction. We extract 95 features per epoch using Welch's method [12, 16] over 0.5–30 Hz: relative band power in four canonical bands (Delta 0.5–4 Hz, Theta 4–8 Hz, Alpha 8–12 Hz, Beta 12–30 Hz) for each of 19 channels ($4 \times 19 = 76$ features), plus total band power per channel (19 features). Relative-power features are normalized by total band power within each epoch. We intentionally use a minimal, standard feature set rather than a disease-optimized one.

Feature transformation. After extraction, pipelines differ only in the transformation stage. In the T-test pipeline, we apply univariate T-test feature selection with family-wise error control at $\alpha = 0.05$ (Bonferroni), then scale the selected features to $[0, 1]$ via Min-Max normalization. In the PCA pipeline, features are Z-score standardized before PCA. Without this, a handful of high-amplitude band-power features would dominate the variance, collapsing 95% of explained variance into just 5–12 components instead of the 24–49 we observe across cohorts. All transformations are fit on training data only and applied unchanged to the corresponding test fold. With datasets prepared under a fixed, minimal pipeline, we now define the experimental designs used to test each trap.

3 Evaluation Design

Feature transforms produce one feature vector per epoch. Evaluation design controls how those vectors are split into train and test sets, and how per-epoch predictions are aggregated into a single subject-level decision.

Metrics and decision unit. Each feature vector is one training or test sample. Predictions are aggregated per subject via majority vote. A subject is predicted as disease if more than half of their epoch predictions are disease. Subject accuracy is the primary metric. Epoch accuracy is diagnostic. It weights subjects by epoch count and can diverge from subject accuracy. Tracking both metrics per fold is what makes Trap 3 visible: when epoch accuracy and subject accuracy rank models differently, the right model selection depends on which metric you report. Reporting and aggregation conventions are described in Appendix B.

Table 2: Experiment configurations. $P = 6$ and $P = 2$ are subject-disjoint LPSO with 50 unique folds. SO is a subject-overlap control for Trap 1, run with 10 seeds.

Experiment ID	Feature pipeline	Split type	P	Unique folds
`T-test_P_6`	T-test	Subject-disjoint (LPSO)	6	50
`PCA_P_6`	PCA	Subject-disjoint (LPSO)	6	50
`T-test_P_2`	T-test	Subject-disjoint (LPSO)	2	50
`PCA_P_2`	PCA	Subject-disjoint (LPSO)	2	50
`T-test_SO`	T-test	Subject-overlap	–	10
`PCA_SO`	PCA	Subject-overlap	–	10

Subject-disjoint splits and LPSO. Subject-disjoint means no subject appears in both train and test within a fold. We implement this with Leave-P-Subjects-Out (LPSO). P subjects are drawn uniformly at random per fold, across 50 randomly generated unique folds at both $P = 6$ and $P = 2$, exposing Trap 2 in two ways: it actively searches for lucky folds where an easy subject draw inflates accuracy, and it measures result stability via IQR across folds. A large IQR means the held-out subjects heavily determine the accuracy. A single-fold study could land anywhere within the IQR's range, or beyond, without knowing the full distribution. Subject-overlap splits are used only as a Trap 1 memorization control [7, 8, 17]. The same design applies identically across all five datasets (Table 2).

Table 3 reports feature dimensionality per dataset and split. T-test feature counts vary fold-to-fold because selection is re-fit on each fold's training data. PCA component counts are comparatively stable across folds.

Table 3: Feature dimensionality per dataset and split. T-test counts vary fold-to-fold because the selected features depend on which subjects are in the training fold. PCA counts are stable across folds. Ranges across 50 folds ($P = 6$, $P = 2$) or 10 seeds (SO).

	T-test features selected			PCA components retained		
Dataset	**P_6**	**P_2**	**SO**	**P_6**	**P_2**	**SO**
AD	74–84	75–83	75	34–35	34–35	34–35
FTD	72–86	76–88	84–86	32–35	33–34	33–34
MDD (eyes-closed)	67–73	69–73	68–70	24–27	25–26	25–26
MDD (eyes-open)	65–75	67–73	68–70	28–31	29–31	29–30
SCZ	45–72	52–67	58–61	46–49	47–48	48

Models and hyperparameters. We evaluate two feature pipelines (T-test and PCA) crossed with four classifiers (KNN, SVM, XGBoost, MLP) and a fixed 3-setting hyperparameter grid per model (12 model$\times$HP configurations per fold; full grid in Appendix E). Implementations use scikit-learn [13] and XGBoost [14]. Total independent fit-transform-predict cycles: 13,200. We first demonstrate all three traps on AD as an anchor cohort before replicating across all five diseases.

4 Results

We demonstrate all three traps on the AD cohort (65 subjects) as an anchor. AD is the right signal strength for this as the T-test accuracy is real but modest (0.70–0.74 subject-disjoint), PCA accuracy is near chance, and all three traps fire clearly in the different scenarios. Trap 1 shows subject-overlap accuracy far above subject-disjoint accuracy. Trap 2 shows that even with clean splits, accuracy is not a fixed number but a range that depends on the subjects in the fold. Trap 3 shows that epoch accuracy, the metric most commonly reported, does not match subject accuracy, which is the metric that matters in clinical settings.

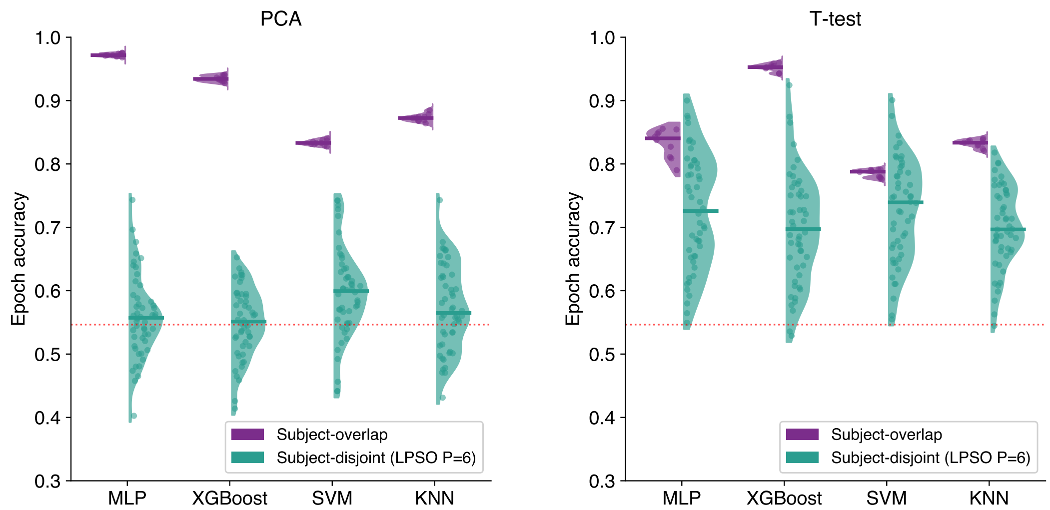

Figure 1: **Trap 1 (AD):** subject-overlap vs. subject-disjoint ($P = 6$, 50 folds), all four models (epoch accuracy). T-test (left), PCA (right). Horizontal bars are medians. Dotted red line marks epoch-level chance accuracy. T-test best-model accuracy drops from 0.96 (subject-overlap) to 0.74 (subject-disjoint).

Trap 1: Identity Leakage. Figure 1 shows subject-overlap accuracy far above subject-disjoint across all models and both pipelines. T-test averages drop from 0.85 (subject-overlap) to 0.72 (subject-disjoint), a 13.6 pp inflation; PCA averages collapse from $\sim$0.93 to $\sim$0.58. Models trained on subject-overlap folds exploit person-specific EEG identity rather than disease signal. A study reporting a single subject-overlapping accuracy near 0.96 would be citing a number that does not exist under honest evaluation.

Trap 2: Lucky Folds. Figure 2 and Table 4 show IQR nearly doubling when reducing P from 6 to 2 while the median barely moves. In both cases, the fold you happen to draw determines your result more than the model you chose. Consider a fold where the 3 control subjects happen to be the youngest and healthiest in the dataset, and the 3 AD subjects have the most advanced disease. Both epoch and subject accuracy will be near perfect — not because the model is good, but because the fold was lucky. T-test's higher baseline variance comes partly from feature selection varying fold-to-fold. PCA component counts are stable across folds (34–35 for AD). T-test selects between 74–84 features at $P = 6$ and 75–83 at $P = 2$ for AD (Table 3). A single-fold study at $P = 2$ could report nearly perfect accuracy and be entirely correct about that fold, and entirely wrong about whether the model generalizes to other folds.

Table 4: **Trap 2 (AD):** Exemplar IQR inflation under cohort-size reduction (epoch accuracy). For both MLP (T-test) and SVM (PCA), reducing P from 6 to 2 roughly doubles the IQR (2.2× and 1.8× respectively) while the median barely moves. The min–max range expands more dramatically as the T-test widens from 0.549–0.900 ($P = 6$) to 0.089–0.975 ($P = 2$), meaning individual lucky folds become more extreme as the test cohort shrinks.

Feature	P	**Model (HP)**	**Median**	**IQR (pp)**	**Min – Max**	**IQR Ratio**
T-test	$P = 6$	MLP (`layers=[100]`)	0.726	14.3	0.549 – 0.900	—
T-test	$P = 2$	MLP (`layers=[100]`)	0.749	31.8	0.089 – 0.975	2.2×
PCA	$P = 6$	SVM (`kernel=rbf`)	0.599	8.3	0.441 – 0.743	—
PCA	$P = 2$	SVM (`kernel=rbf`)	0.550	14.8	0.247 – 0.786	1.8×

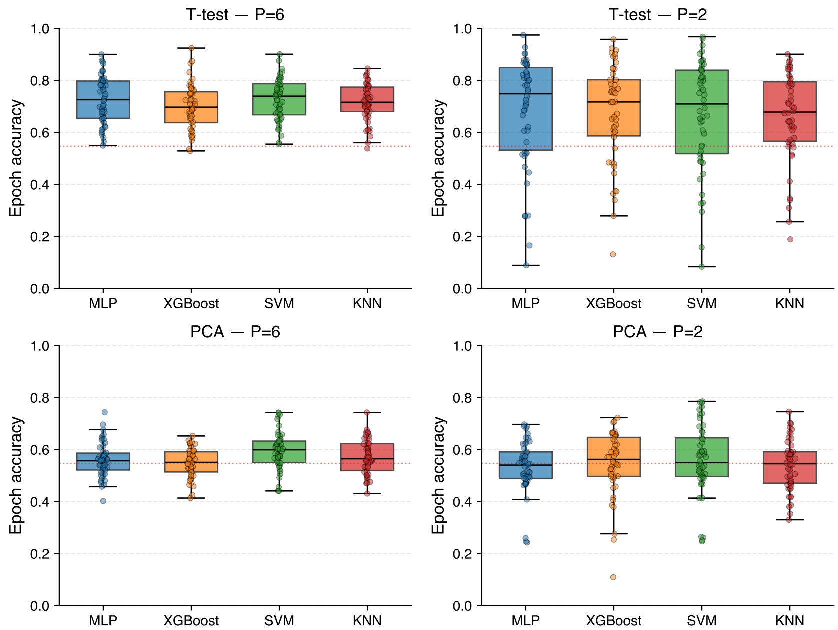

Figure 2: **Trap 2 (AD):** each dot is one of the 50 unique folds run per configuration, $P = 6$ (left) vs. $P = 2$ (right) (epoch accuracy). PCA (top), T-test (bottom). Dotted red line marks epoch-level chance accuracy. T-test IQR grows from 14.3 pp ($P = 6$) to 31.8 pp ($P = 2$).

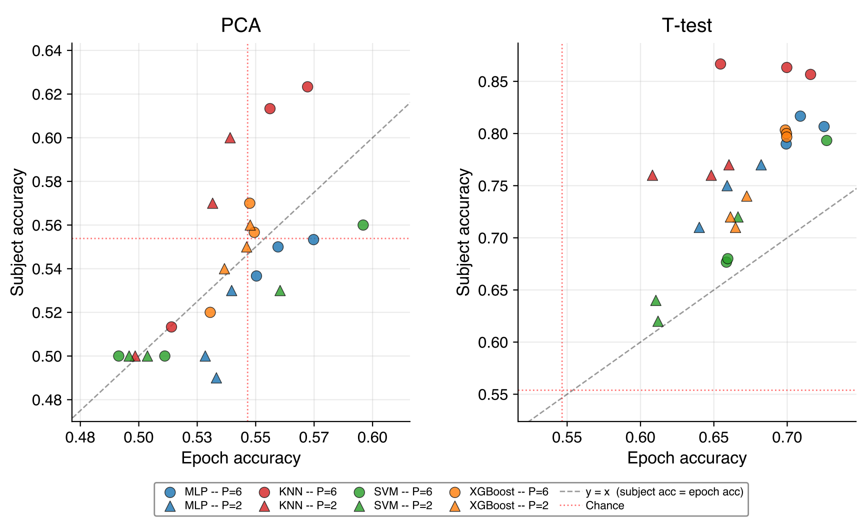

Figure 3: **Trap 3 (AD):** each point is one fold, $P = 6$ (circles) vs. $P = 2$ (triangles). PCA (left), T-test (right). X-axis: epoch accuracy; Y-axis: subject accuracy. Diagonal $y = x$ marks equality. Red dotted lines mark epoch-level chance (vertical) and subject-level chance (horizontal). PCA clusters at subject accuracy ≈ 0.50. In both PCA and T-test, KNN ranks last by epoch accuracy yet first by subject accuracy.

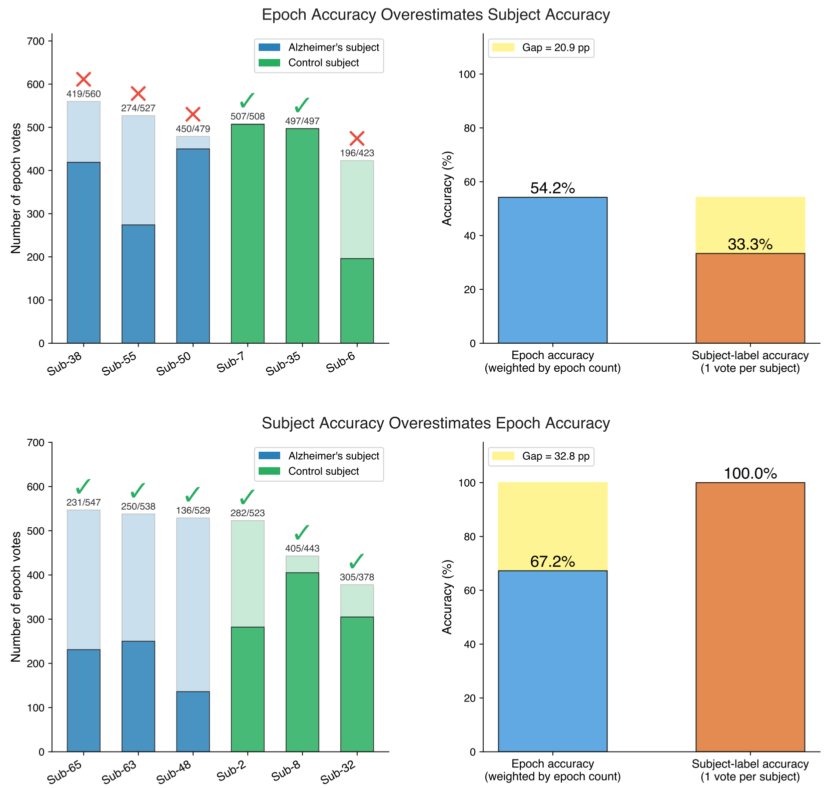

Figure 4: **Trap 3 (AD) — mismatch mechanism.** Top: SVM (kernel=rbf, C=0.1), T-test, fold {sub-6/7/35/38/50/55} — epoch accuracy 0.542 vs. subject accuracy 0.333; epoch accuracy overestimates subject accuracy by 20.9 pp. Bottom: KNN k=7, T-test, fold {sub-2/8/32/48/63/65} — epoch accuracy 0.672 vs. subject accuracy 1.000; epoch accuracy underestimates subject accuracy by 32.8 pp. Mismatch runs both ways.

Trap 3: Objective Mismatch. Figure 3 shows PCA collapsing to ≈ 0.50 subject accuracy despite positive epoch accuracy. The signal appears real at the epoch level but vanishes at the subject level. In both pipelines, KNN ranks last by epoch accuracy yet first by subject accuracy. Figure 4 shows two concrete examples: SVM overestimates subject accuracy by 20.9 pp because one subject with many epochs dominates the count; KNN underestimates by 32.8 pp because slim majority votes accumulate losing epochs. Selecting by epoch accuracy alone leads to the wrong model. Table 8 (Appendix C) summarises four AD validation configurations. Subject-overlap (80/20). LPSO $P = 6$ with the best pipeline (T-test, MLP). LPSO $P = 6$ with the worst (PCA, KNN). LPSO $P = 2$ (T-test, MLP). Together they expose all three traps in one cohort. Trap 1: subject-overlap inflates epoch accuracy by 11.2 pp (83.8% vs. 72.6%). Trap 3: PCA-KNN at LPSO $P = 6$ collapses to 50.0% subject accuracy despite 56.5% epoch accuracy. Trap 2: at LPSO $P = 2$ the epoch-accuracy IQR doubles ($14.3 \rightarrow 31.8$ pp), and subject accuracy degenerates to 100% with a 50 pp IQR. The AD results establish that all three traps fire under controlled conditions. We now test whether they replicate across four additional diseases with different biomarker profiles, recording conditions, and subject counts.

5 Replication Across FTD, MDD, and SCZ

The five cohorts differ in nearly every way that matters. AD and FTD are neurodegenerative; MDD and SCZ are psychiatric. Recording hardware, sampling rates, reference schemes, and acquisition sites differ across all three source datasets. Subject-disjoint accuracy ranges from 0.57 (SCZ) to 0.81 (MDD EC). Despite these differences, all three traps replicate across every cohort under identical pipelines. Full per-cohort figures are in Appendix J.

Epoch accuracy and subject accuracy are correlated across diseases and models — both reflect underlying signal strength — but diverge in ways that matter for model selection and clinical reporting (Trap 3). For simplicity, Traps 1 and 2 are demonstrated using epoch accuracy, which is the metric the existing literature reports; subject accuracy shows equivalent patterns, summarised in Appendix Tables 15 and 16. All subject-level diagnostic claims in this paper are stated in terms of subject accuracy.

Table 5: **Trap 1 across diseases:** subject-disjoint vs. subject-overlap epoch accuracy (T-test pipeline, avg across 4 models). Subject-overlap is consistently far above subject-disjoint for every disease. Full per-model breakdown in Appendix Table 12; subject accuracy equivalent in Appendix Table 15.

Dataset	**Epoch acc. $P = 6$ (disj.)**	**Epoch acc. overlap**	**Inflation (pp)**
AD	0.715	0.851	+13.6 pp
FTD	0.627	0.871	+24.3 pp
MDD EC	0.814	0.914	+10.0 pp
MDD EO	0.781	0.911	+13.0 pp
SCZ	0.573	0.851	+27.8 pp

Table 5 shows inflation in every disease without exception. The range is $+10.0$ pp (MDD EC) to $+27.8$ pp (SCZ). No disease is immune regardless of biomarker strength. A study on these cohorts using subject-overlapping folds could report inflated accuracy by at least 10 pp and up to 28 pp.

Table 6: **Trap 2 across diseases:** fold-to-fold IQR (epoch accuracy) and maximum single-fold accuracy at both metrics, $P = 6$ and $P = 2$ (T-test pipeline, avg IQR across 4 models, 50 unique folds). IQR is already 8–14 pp at $P = 6$. At $P = 2$ it roughly doubles in every disease. Lucky folds can reach perfect accuracy at both test sizes. Full per-model breakdown in Appendix Table 13; subject accuracy IQR equivalent in Appendix Table 16.

		Ep. Acc. IQR (pp)		**Max Ep. Acc.**		**Max Sub. Acc.**	
Dataset	**IQR ratio**	$P = 6$	$P = 2$	$P = 6$	$P = 2$	$P = 6$	$P = 2$
AD	2.2×	11.8	26.2	0.924	0.975	1.000	1.000
FTD	1.8×	11.3	20.4	0.856	0.973	1.000	1.000
MDD EC	2.2×	14.1	30.6	0.996	1.000	1.000	1.000
MDD EO	1.9×	13.8	26.8	0.989	1.000	1.000	1.000
SCZ	2.1×	8.7	17.9	0.788	0.834	1.000	1.000

Table 6 shows fold variance is already substantial at $P = 6$ across every disease (8–14 pp IQR), meaning a single fold is insufficient even before reducing to $P = 2$. At $P = 2$, IQR roughly doubles in every disease. MDD EC and MDD EO reach a maximum fold epoch accuracy of 1.00 at $P = 2$; AD reaches 0.97 and FTD 0.97, showing that even epoch-level accuracy can be lucky-fold inflated near the ceiling. The subject accuracy story is even starker: every disease has at least one $P = 6$ fold where the model classifies all six held-out subjects correctly.

Table 7: **Trap 3 across diseases:** epoch accuracy vs. subject accuracy (T-test pipeline, $P = 6$, avg across 4 models). On average across models, subject accuracy exceeds epoch accuracy in every disease without exception. Full T-test and PCA results at $P = 6$ and $P = 2$ in Appendix Table 14.

Dataset	**Avg epoch acc.**	**Avg subject acc.**	**Avg gap (pp)**
AD	0.715	0.833	+11.9 pp
FTD	0.627	0.667	+3.9 pp
MDD EC	0.814	0.854	+4.1 pp
MDD EO	0.781	0.833	+5.2 pp
SCZ	0.573	0.667	+9.4 pp

Table 7 shows subject accuracy exceeding epoch accuracy in every disease. The gap ranges from $+3.9$ pp (FTD) to $+11.9$ pp (AD). PCA collapses to near-chance subject accuracy across all diseases despite positive epoch accuracy. On average subject accuracy exceeds epoch accuracy, but individual models diverge. Selecting by epoch accuracy alone could pick a model that performs worse at the subject level, as Figure 4 shows.

The direction and magnitude of the gap depend on model architecture and cohort. In aggregate, subject accuracy exceeds epoch accuracy across all five diseases. But individual models within the same disease can go in opposite directions. What is consistent is not the direction but the instability. Epoch accuracy does not reliably predict subject accuracy. The model that looks best by one metric can look worst by the other. Full per-model breakdowns are in Appendix K.

Across five diseases spanning two diagnostic categories, three independent datasets, and varying ranges of accuracy, all three traps fire without exception. The traps are not properties of AD, of EEG, or of any particular biomarker. They are structural properties of evaluation design, and could present wherever small-cohort clinical disease classification is conducted.

6 Discussion

Three evaluation design choices conspire to inflate reported EEG classification accuracy, and likely do the same across any clinical ML setting with small subject pools. Identity leakage allows models to memorize person-specific structure rather than disease patterns (Trap 1). Small held-out cohorts create large fold-to-fold variability that invites undetected cherry-picking (Trap 2). And the metric most commonly reported, epoch accuracy, does not necessarily match the clinical decision unit, leading to mis-ranking models (Trap 3).

The replication of these results across different EEG datasets confirms that all three traps operate across diseases with different biomarker profiles. MDD, with the strongest genuine cross-subject signal (≈ 0.83 T-test LPSO median), still exhibits overlap inflation, lucky-fold maxima of 1.00, PCA subject-accuracy collapse, and epoch-to-subject accuracy mismatch (Table 7) with equal or greater severity than weaker-signal diseases. At the other extreme, SCZ with the weakest disjoint signal (0.56–0.60) still shows overlap inflation of $+19$ to $+32$ pp, IQR inflation of 1.7–$2.5\times$, and an epoch-to-subject accuracy gap of $+9.4$ pp. These traps are structural properties of evaluation design, not of the biomarker.

The pipelines here are deliberately minimal. No task-specific feature engineering, no deep learning, no ensembling. Despite this, MDD achieves subject accuracy of 0.83 under rigorous LPSO evaluation. Genuine cross-subject neurophysiological signal exists and is detectable under honest evaluation. More carefully optimised systems may achieve much higher accuracy which are genuine. The present results are a lower bound, not a ceiling.

Reporting standard. These three traps jointly imply a minimum reporting standard for any study making subject-level diagnostic claims. Concretely, for subject-independent EEG claims: **(i) use subject-disjoint splits** (addresses Trap 1); **(ii) quantify cohort sensitivity** and report distributions — median and IQR over many folds, not a single value; repeated resampling (LPSO) is one way to achieve this, but Bayesian approaches or analytic uncertainty estimates over cohort composition are equally valid alternatives (addresses Trap 2); **(iii) report subject accuracy** since subject accuracy is the primary metric when the claim concerns subject-level classification (addresses Trap 3); and **(iv) publish the split-generation method and split manifests listing the held-out test subject IDs per fold** (see Appendix D) so results are reproducible and comparable. This emphasis on transparent methodology is aligned with prior EEG evaluation recommendations and leakage-focused work more broadly [11, 3, 9].

Guidance for LOSO designs. LOSO tests every subject exactly once. There is no resampling and no fold variance to report. But the underlying problem does not go away — it just becomes invisible.

Two simple diagnostics make it visible again, and both require zero additional experiments.

First, report a confusion matrix. Show how many disease subjects were correctly classified and how many control subjects were correctly classified separately. Do not collapse these into a single accuracy number. A model that gets 90% overall accuracy but misses half the disease cases is clinically useless. A confusion matrix shows this. A single number does not.

Second, if the model operates on epochs, report epoch accuracy as mean $\pm$ SD across subjects within each class. A large SD indicates high between-subject heterogeneity — some subjects are substantially easier to classify than others. This means the reported mean accuracy is an unstable estimate, sensitive to which specific subjects happened to be recruited. This is the LOSO analogue of a lucky fold: where LPSO resampling reveals instability through variance across random splits, per-subject SD reveals the same instability through variance across individual subjects.

7 Conclusion

EEG disease classification can appear stronger than it really is when evaluation is too loose. We identified three traps: subject overlap inflates accuracy, small held-out cohorts create lucky folds, and epoch accuracy can mis-rank models relative to subject accuracy. The practical guidance is straightforward: use subject-disjoint evaluation, report cohort sensitivity across folds, report subject accuracy when the claim is subject-level, and publish the split-generation method and split manifests listing the held-out test subject IDs per fold. These practices should help readers judge which results are reliable and help authors design experiments that avoid the same traps. The pipelines here are intentionally minimal, the underlying biomarker signal is real, and future work should test how these traps behave in other domains and larger cohorts.

Limitations and Future Work. All five datasets use resting-state EEG with the same 95 relative-band-power features; replication on task-state recordings, dense-array setups, or deep feature extractors is needed to assess how the traps scale with model capacity. MDD uses only one recording protocol; additional cohorts are needed. SCZ has only 28 subjects, limiting statistical precision at the subject level. The traps may manifest with different magnitudes in very large cohorts ($N \gg 100$) where P can be increased substantially, as in recent foundation-model benchmarks [21]; how the traps scale with cohort size remains an open empirical question. The three traps arise from structural properties of evaluation design, not from properties of EEG. Small subject pools, repeated within-subject measurements, and aggregation across observations are generally mechanisms for neuroimaging and clinical ML. Domains sharing these properties may face the same traps. We leave empirical replication in those domains to future work.

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Code and Data Availability

All split manifests, YAML pipeline configurations, evaluation source code, and run-level results are available at `https://github.com/anon333science2026-afk/eeg-evaluation-traps`. The repository contains a README describing the contents.

A Dataset Curation and BIDS Packaging

All three datasets use the 19-channel International 10–20 electrode set but differ in sampling frequency, recording duration, and reference scheme.

AD (OpenNeuro ds004504). Eyes-closed resting-state EEG acquired at 500 Hz. Referential montage with Cz reference. Recording durations 5.1–21.3 min (AD) and 12.5–16.5 min (controls). Arrived in BIDS format [1, 15].

FTD (OpenNeuro ds004904). Eyes-closed resting-state EEG acquired at 500 Hz. Same acquisition protocol as the AD dataset. Arrived in BIDS format [1, 15].

MDD (Mumtaz et al. 2017). Both eyes-open and eyes-closed conditions acquired at 256 Hz. Required curation from a flat EDF archive: inconsistent filenames were normalized, exact-copy duplicates and suspected cross-subject duplicate pairs were removed, and task recordings were excluded, retaining 115 rest recordings from 61 unique subjects [22]. Fully reproducible from raw files via a one-command Python builder.

SCZ (Olejarczyk & Jernajczyk 2017). Resting-state EEG acquired at 250 Hz. Reference between Fz and Cz. Recording durations 12.3–36.2 min (SCZ) and 14.4–18.6 min (controls). Eye-state is not distinguished in source metadata. Required only channel reordering for one recording (`h12.edf`) during BIDS packaging [23]. Fully reproducible from raw files via a one-command Python builder.

B Reporting Conventions and Pooling over Runs

All epoch-accuracy statistics reported in this manuscript are derived from the `test_accuracy` column in `raw_info/all_experiments_combined.csv`. Unless otherwise stated, subject-independent LPSO *epoch-accuracy* headline values follow **Median(all runs)**: the median and IQR are computed over *all* recorded runs for the relevant (experiment, model), pooling across folds *and* hyperparameter configurations.

Summary-table aggregation rule. All summary tables in §5 and all result figures use a single aggregation rule: for each model we pick the hyperparameter configuration with the highest median test accuracy across folds, take the median of that configuration's 50 fold accuracies, and average across the 4 models. KNN is locked to $k = 7$ in all summary tables and figures for cross-cohort consistency and to avoid the degenerate $k = 1$ nearest-neighbour solution.

C AD Performance Summary

Table 8: Performance summary (AD). Subject-accuracy is the primary subject-independent metric; epoch accuracy is diagnostic.

Validation method	Leak.	Primary subject acc.	Diagnostic epoch acc.	Interp.
Subject-dependent 80/20 (AD)	Yes	–	83.8% (IQR 8.6 pp)	Inflated by subject-overlap
LPSO ($P = 6$; T-test; MLP)	No	83.33% (IQR 16.67 pp)	72.6% (IQR 14.3 pp)	Subject-independent (primary)
LPSO ($P = 6$; PCA; KNN)	No	50.00% (IQR 33.33 pp)	56.5% (IQR 10.4 pp)	Near-baseline at subject level
LPSO ($P = 2$; T-test; MLP)	No	100.00% (IQR 50.00 pp)	74.9% (IQR 31.8 pp)	Highly unstable; heavily quantized

D Split Manifests (held-out subjects per fold)

Table 9 lists the actual held-out test subjects for the first six $P = 6$ folds of the AD LPSO run, sorted by lowest subject index. Full manifests for all five cohorts (all 50 $P = 6$ folds and all 50 $P = 2$ folds per cohort) are available in the repository. To make subject-independent results reproducible and comparable, we provide split manifests listing the held-out test subject IDs per fold for the LPSO settings used in this paper.

Table 9: Held-out test subject IDs for the first six $P = 6$ folds (AD, LPSO Random-50). These are the actual subject sets used by every result reported for AD at $P = 6$; full manifests for all 50 $P = 6$ folds and all 50 $P = 2$ folds per cohort are in the repository.

Setting	**Held-out subject IDs**
$P = 6$ (fold 1)	1, 6, 11, 50, 59, 64
$P = 6$ (fold 2)	1, 13, 29, 40, 45, 53
$P = 6$ (fold 3)	1, 16, 30, 49, 55, 56
$P = 6$ (fold 4)	1, 33, 35, 38, 40, 61
$P = 6$ (fold 5)	2, 11, 19, 41, 62, 63
$P = 6$ (fold 6)	3, 26, 30, 40, 52, 63

E Hyperparameter Grid

- **KNN:** metric = euclidean; weights = uniform; $k \in \{1, 7, 15\}$.
- **SVM:** $C = 0.1$; $\gamma = \texttt{auto}$; kernel $\in \{\texttt{linear}, \texttt{rbf}, \texttt{poly}\}$.
- **MLP:** activation = tanh; $\alpha = 0.1$; hidden layer sizes $\in \{(100), (150, 50), (200, 100, 50)\}$.
- **XGBoost:** n_estimators = 100; learning_rate = 0.2; subsample = 0.7; max_depth $\in \{3, 6, 9\}$.

F Supplementary: Biomarker Discovery, AD

The T-test pipeline recovers known AD biomarkers: posterior Alpha decrease and Delta increase, consistent with established EEG markers of Alzheimer's disease. The alignment between statistically discriminative features and known disease physiology confirms that the T-test pipeline is detecting genuine neurophysiological signal rather than noise. This is a point that holds even as all three evaluation traps still fire.

Table 10: Top 10 EEG biomarkers for AD vs control classification (Cohen's $|d|$; 800 μV rejection threshold, 34,044 epochs).

Feature	**Mean AD**	**Mean Control**	**Ratio**	**Cohen's d**
O2 × Alpha	0.0400	0.1064	2.66	0.822
T5 × Alpha	0.0364	0.0916	2.51	0.739
O1 × Alpha	0.0389	0.0951	2.45	0.725
T6 × Alpha	0.0379	0.0772	2.04	0.617
O2 × Delta	0.8888	0.8260	1.08	0.616
P3 × Alpha	0.0278	0.0538	1.94	0.547
O1 × Delta	0.8898	0.8378	1.06	0.521
T5 × Delta	0.8896	0.8389	1.06	0.512
Pz × Alpha	0.0259	0.0480	1.86	0.508
P4 × Alpha	0.0285	0.0470	1.65	0.434

G Supplementary: T-Test Top Features Across Diseases

Figure 5 shows top-10 features by Cohen's $|d|$ for each disease. AD and FTD show occipital/temporal Alpha decreases; MDD shows widespread Beta elevation ($|d| > 1.0$, substantially larger than other diseases); SCZ shows frontal/temporal Alpha lateralization. These disease-specific signatures confirm the T-test pipeline is detecting genuine neurophysiological signal rather than noise.

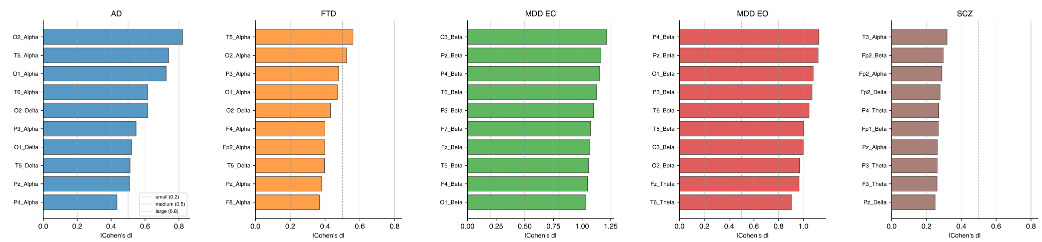

Figure 5: Top-10 EEG features by |Cohen's d| for each disease. AD and FTD show occipital/temporal Alpha decreases and Delta increases. MDD shows widespread Beta elevation ($|d| > 1.0$).

Table 11: Top-5 T-test features per disease (by |Cohen's d|; ↑/↓ = direction in disease vs. control).

Rank	AD	FTD	MDD EC	MDD EO	SCZ
1	O2_Alpha ↓ (0.82)	T5_Alpha ↓ (0.56)	C3_Beta ↑ (1.22)	P4_Beta ↑ (1.12)	T3_Alpha ↑ (0.32)
2	T5_Alpha ↓ (0.74)	O2_Alpha ↓ (0.52)	Pz_Beta ↑ (1.16)	Pz_Beta ↑ (1.12)	Fp2_Beta ↑ (0.30)
3	O1_Alpha ↓ (0.72)	P3_Alpha ↓ (0.48)	P4_Beta ↑ (1.15)	O1_Beta ↑ (1.08)	Fp2_Alpha ↑ (0.29)
4	T6_Alpha ↓ (0.62)	O1_Alpha ↓ (0.47)	T6_Beta ↑ (1.13)	P3_Beta ↑ (1.07)	Fp2_Delta ↓ (0.28)
5	O2_Delta ↑ (0.62)	O2_Delta ↑ (0.43)	P3_Beta ↑ (1.10)	T6_Beta ↑ (1.05)	P4_Theta ↓ (0.27)

H Pipeline Architecture (PySpark + Ray)

The leakage-free constraint — that every data-dependent step be fitted on training data only, independently per fold — makes naive sequential execution prohibitively slow at 13,200 runs. To meet this constraint at scale, experiments were executed via a distributed PySpark/Ray pipeline; the Spark container handles preprocessing and feature extraction while the Ray container orchestrates group-aware subject splits, model training, and per-fold evaluation. Total independent fit-transform-predict cycles: LPSO $P = 6$ (5 datasets $\times$ 2 pipelines $\times$ 50 folds $\times$ 12 HP = 6,000) + LPSO $P = 2$ (6,000) + subject-overlap controls (1,200) = **13,200 runs**. The pipeline and run results are available in the repository. Replication of the reported results does not require the pipeline: the split manifests, YAML configurations, and evaluation code are sufficient to reproduce any individual result.

Compute resources. All jobs ran on a shared HPC cluster via SLURM, with per-job allocations specified in each YAML config:

- Build (Singularity image): 2 CPUs, 12 GB RAM, $\leq$ 15 min walltime
- PySpark stage (preprocess + features): 12 CPUs, 56 GB RAM, $\leq$ 8 h walltime
- Ray stage (ML grid search): 12 CPUs, 56 GB RAM, $\leq$ 8 h walltime

LPSO and subject-overlap (W_C) jobs request the same allocation. End-to-end wall time per job was approximately 15 minutes for an LPSO configuration (50 folds $\times$ 12 model–HP combinations) and approximately 4 minutes for a W_C configuration. Total compute: 120 SLURM jobs (20 LPSO + 100 W_C) consuming $\sim$170 CPU-hours across all 13,200 fit–transform–predict cycles, or roughly 38 CPU-seconds per fit. Hardware: commodity x86 CPU nodes; no GPU was used.

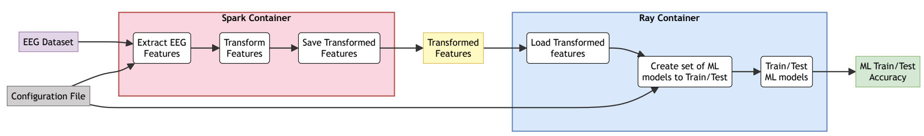

Figure 6: Distributed EEG analysis pipeline (PySpark + Ray). The Spark container handles preprocessing and feature extraction; the Ray container orchestrates group-aware subject splits, model training, and per-fold testing.

I Example Configuration

```yaml
project:
  name: T-test_L_6_C
  output_dir: ./data
  experiment_type: ML Classification
  subjects_or_events: subjects
  deployment_method: Singularity with Slurm
  random_seed: 42

data_input:
  groups:
    alz:
      - /path/to/sub-001/eeg/sub-001_task-eyesclosed_eeg.set
    cntrl:
      - /path/to/sub-037/eeg/sub-037_task-eyesclosed_eeg.set

preprocessing:
  bands:
    Delta: [0.5, 4]
    Theta: [4, 8]
    Alpha: [8, 12]
    Beta: [12, 30]
  window_size: 3.0
  sliding_window: 0.5
  normalize_psd: 'Yes'
  epoch_rejection:
    reject: 800.0
    flat: 0.3

feature_transformation:
  transformations:
    - Univariate T-test
    - MinMax scaler
  f_test_selection_mode: fwe
  f_test_selection_threshold: 0.05

data_transformation_strategy:
  strategy: LPSO (Leave-P-Subjects-Out)
  lpso_subjects_per_group: 6
```

Listing 1: Example YAML config: T-test pipeline, LPSO $P = 6$

J Supplementary: Replication Across FTD, MDD, and SCZ

Each disease is shown in sequence: Trap 1 (subject-overlap vs. disjoint), Trap 2 (fold-level IQR at $P = 6$ and $P = 2$), then Trap 3 (epoch vs. subject accuracy).

FTD (52 subjects)

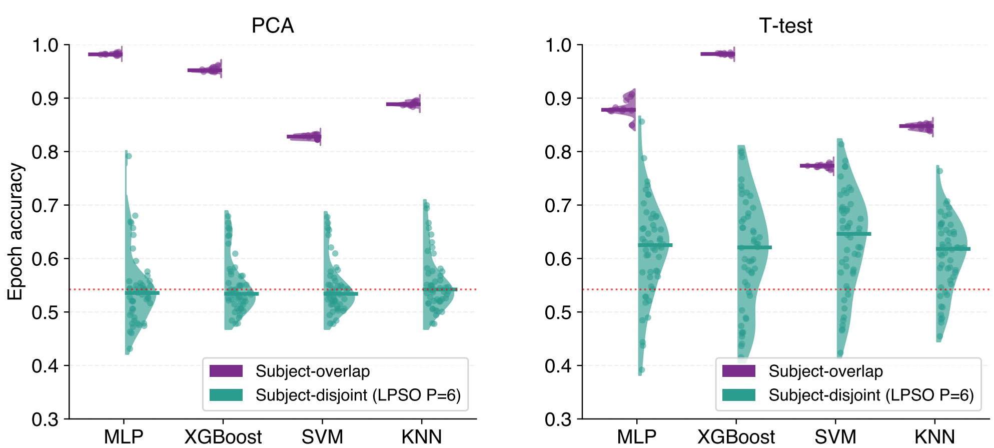

Figure 7: **Trap 1 (FTD)** (epoch accuracy): T-test disjoint medians 0.62–0.65; overlap inflates to 0.77–0.98. PCA disjoint near chance; overlap inflates +29 to +45 pp.

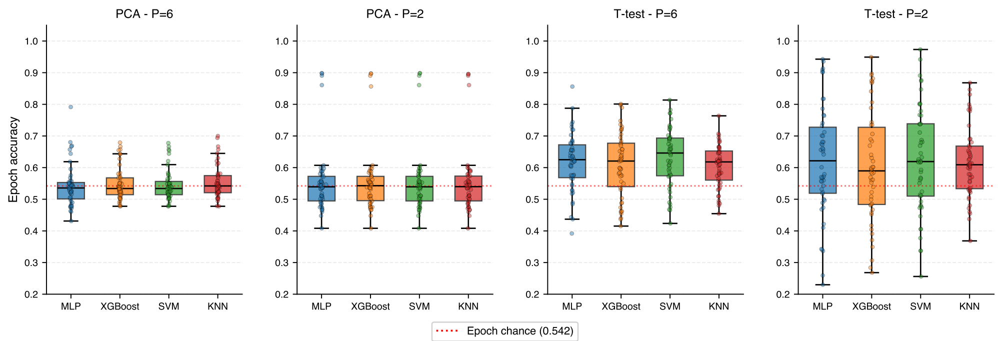

Figure 8: **Trap 2 (FTD)** (epoch accuracy): T-test $P = 6$ IQR 9–14 pp; at $P = 2$ IQR inflates 1.5–$2.0\times$ and lucky-fold epoch accuracy maxima reach 0.87–0.97.

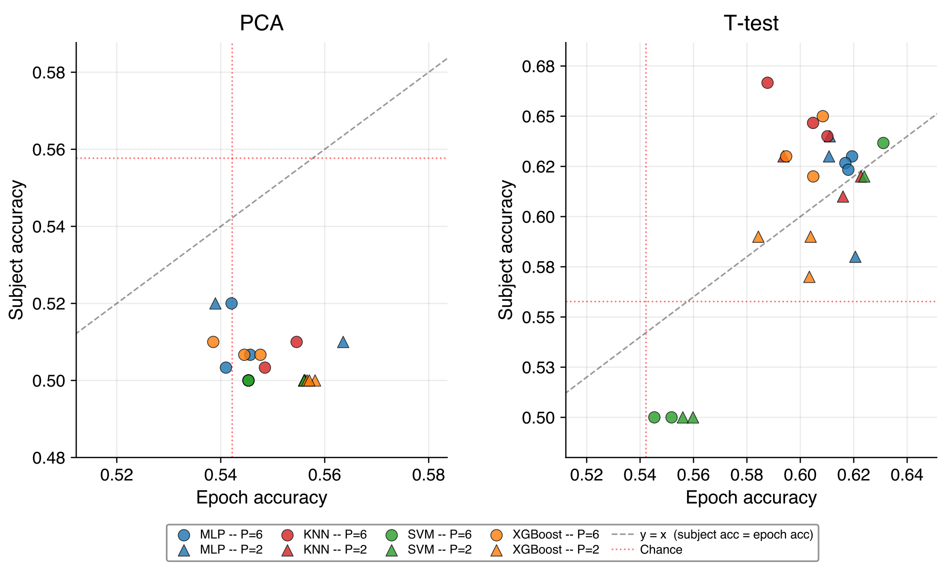

Figure 9: **Trap 3 (FTD):** T-test subject accuracy (0.58–0.67 at $P = 6$) consistently above epoch accuracy. PCA collapses to subject accuracy ≈ 0.50.

MDD (56 EC / 59 EO subjects)

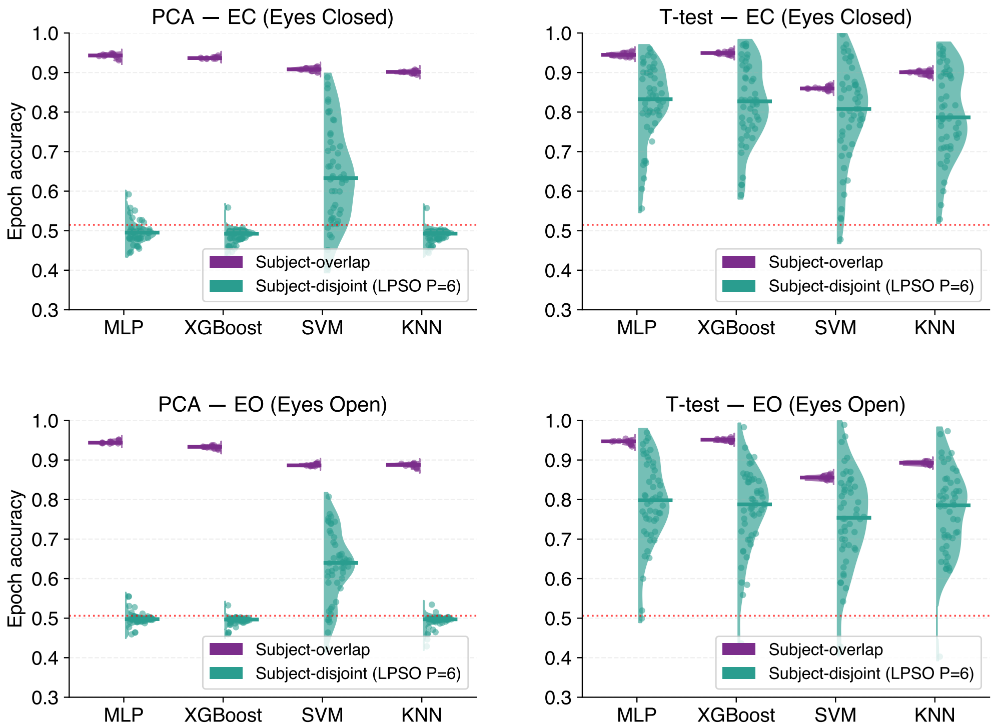

Figure 10: **Trap 1 (MDD)** (epoch accuracy): Top row = eyes-closed (EC), bottom row = eyes-open (EO). T-test disjoint genuinely strong (0.79–0.83 EC; 0.75–0.80 EO); overlap still inflates by +5 to +16 pp. PCA inflates +25 to +45 pp.

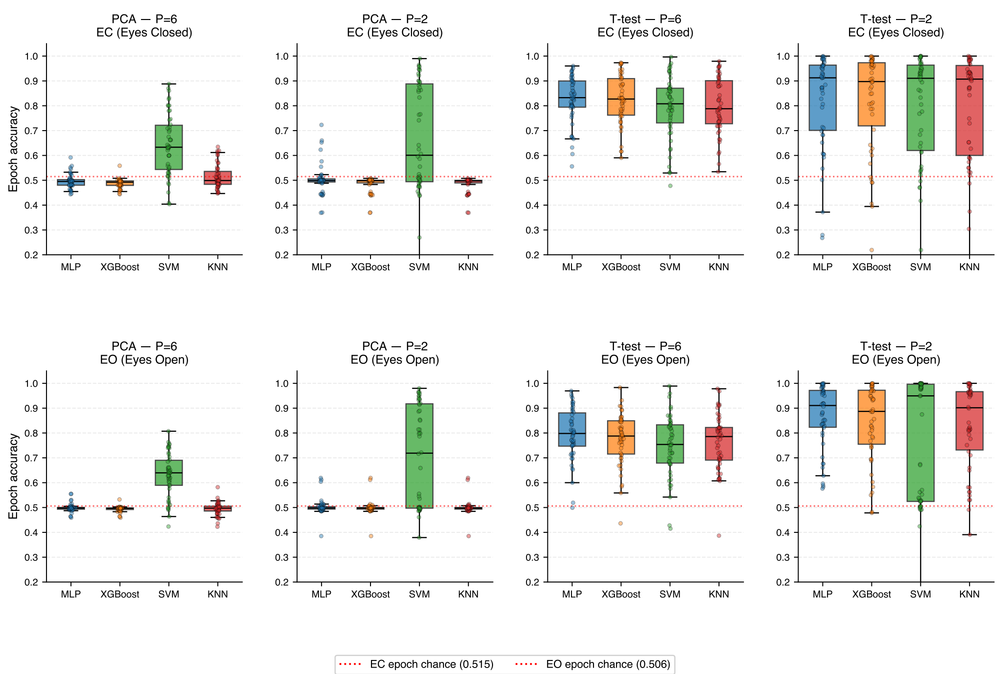

Figure 11: **Trap 2 (MDD)** (epoch accuracy): Top row = EC, bottom row = EO. T-test $P = 2$ folds reach 1.00 for all models in both conditions; EC IQR inflates 1.5–2.6×, EO 1.1–3.1× (SVM 47.2 pp).

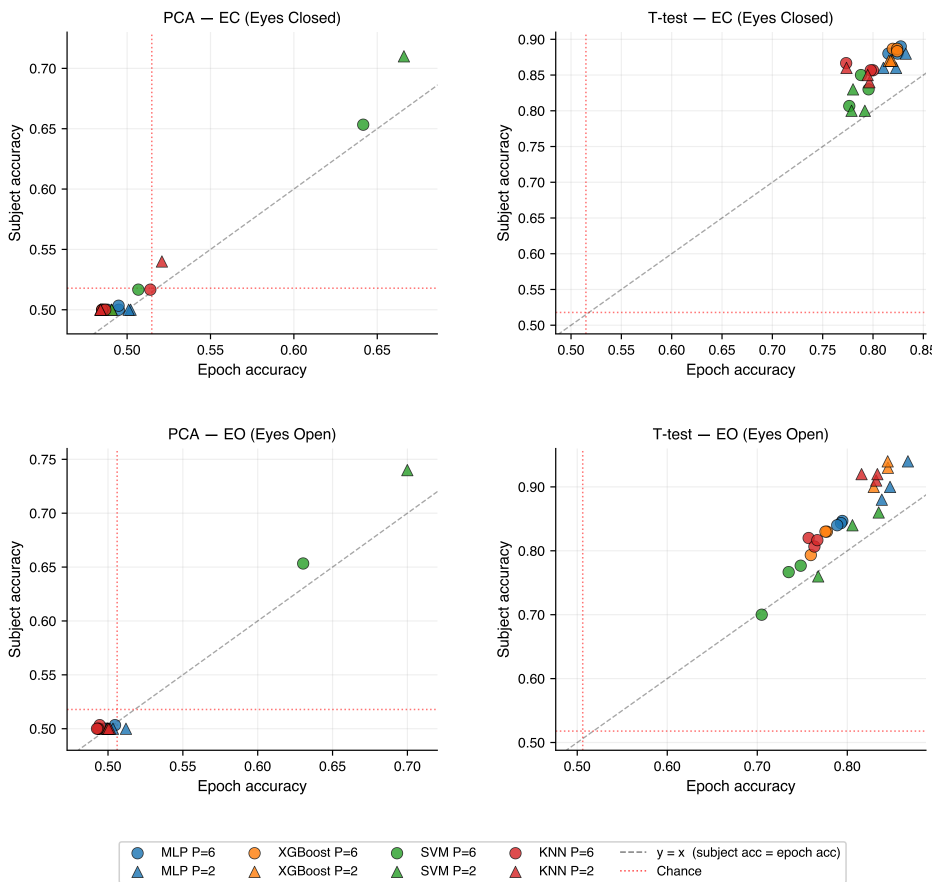

Figure 12: **Trap 3 (MDD):** T-test subject accuracy (0.77–0.89 at $P = 6$) consistently tracks or exceeds epoch accuracy. PCA collapses to subject accuracy ≈ 0.50 in both conditions.

SCZ (28 subjects)

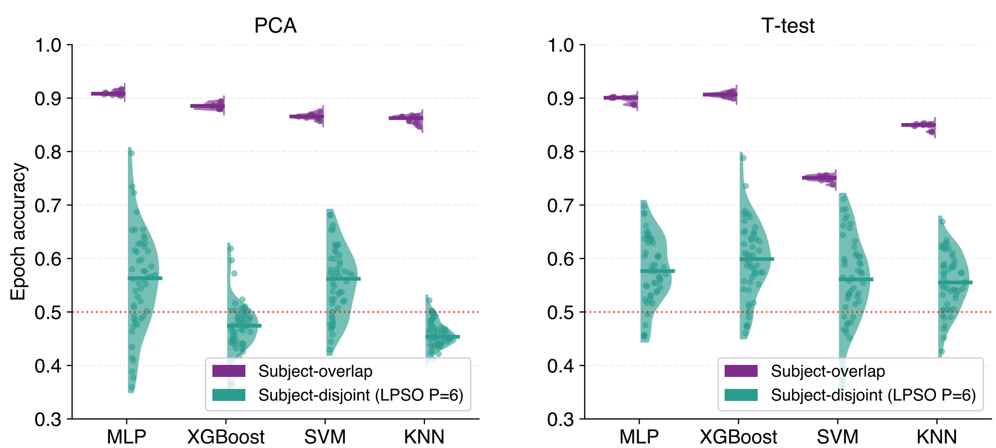

Figure 13: **Trap 1 (SCZ)** (epoch accuracy): T-test disjoint medians 0.56–0.60; overlap inflates to 0.75–0.91 (+19 to +32 pp). PCA inflates +30 to +41 pp.

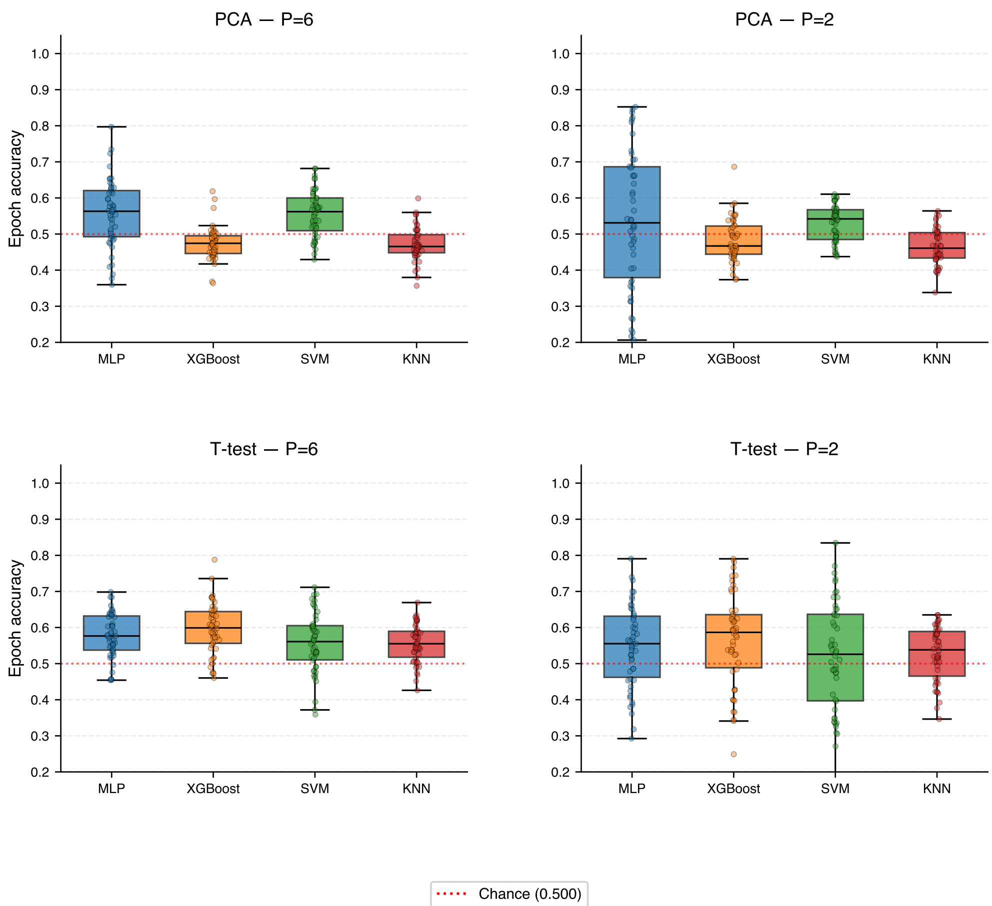

Figure 14: **Trap 2 (SCZ)** (epoch accuracy): T-test $P = 6$ IQR 7–10 pp; at $P = 2$ IQR inflates 1.7–2.5× and lucky-fold epoch accuracy maximum reaches 0.834.

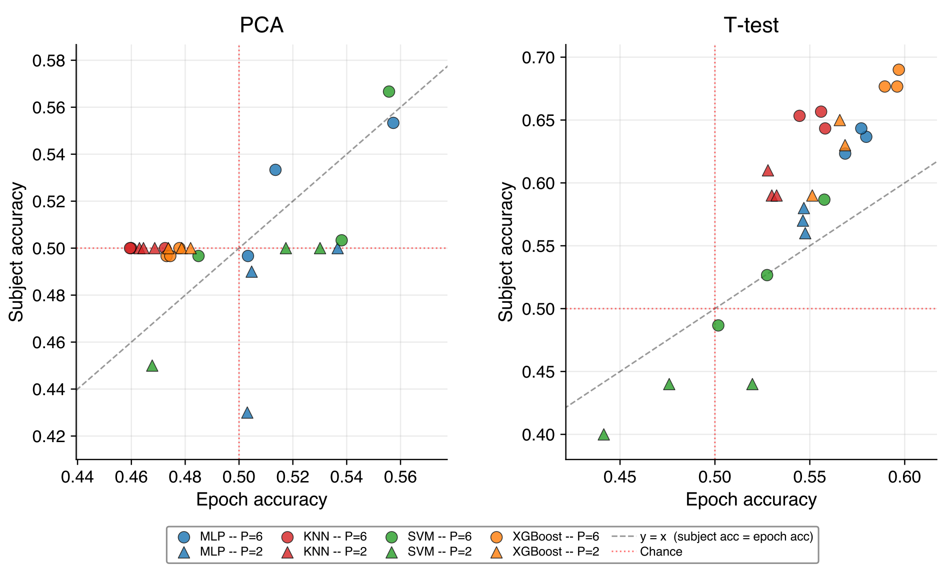

Figure 15: **Trap 3 (SCZ):** T-test subject accuracy (0.59–0.68 at $P = 6$) above epoch accuracy. PCA near chance; some $P = 2$ configurations drop below chance.

K Full Replication Tables

Tables 12–14 report the full per-model, per-pipeline breakdown of all three traps across all five diseases. These complement the summary tables in the main body (Tables 5–7) by showing the min-max range across all four models rather than the average.

Table 12: Appendix A — Trap 1: Identity Leakage (epoch accuracy). T-test and PCA pipelines (all models, best HP). Inflation = SO minus LPSO $P = 6$ per model; Min/Max is range across 4 models.

	T-test $P = 6$ (disj.)		**T-test infl. (pp)**		**PCA $P = 6$ (disj.)**		**PCA infl. (pp)**	
Dataset	Min	Max	Min	Max	Min	Max	Min	Max
AD	0.70	0.74	+5	+26	0.55	0.60	+23	+41
FTD	0.62	0.65	+13	+36	0.53	0.54	+29	+45
MDD EC	0.79	0.83	+5	+12	0.49	0.63	+28	+45
MDD EO	0.75	0.80	+10	+16	0.50	0.64	+25	+45
SCZ	0.56	0.60	+19	+32	0.45	0.56	+30	+41

Table 13: Appendix B — Trap 2: Lucky Folds (epoch accuracy IQR inflation). T-test and PCA pipelines (50 unique folds; all models, best HP). Min/Max is range across 4 models. Note: MDD PCA IQR ratios are erratic (some $< 1\times$) because PCA accuracy on MDD is near-chance (0.49–0.64); IQRs reflect noise rather than fold variance.

	T-test $P = 6$ IQR		**T-test $P = 2$ IQR**		**T-test ratio**		**PCA $P = 6$ IQR**		**PCA $P = 2$ IQR**		**PCA ratio**	
Dataset	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
AD	9.0	14.3	19.2	32.1	1.8×	2.7×	6.5	10.4	8.4	15.0	0.8×	1.9×
FTD	9.2	13.7	13.5	24.4	1.5×	2.0×	4.2	5.4	7.7	7.8	1.4×	1.9×
MDD EC	10.5	17.8	25.4	34.9	1.7×	2.5×	1.8	17.8	1.0	39.4	0.6×	2.2×
MDD EO	13.4	15.4	14.8	47.2	1.1×	3.1×	0.6	10.1	0.6	42.0	1.0×	4.2×
SCZ	7.2	9.5	14.7	24.0	1.7×	2.5×	2.9	12.8	7.0	30.7	0.9×	2.4×

Biomarker features and per-subject heterogeneity

The top T-test-selected features by Cohen's $|d|$ recover known disease-specific physiology: posterior Alpha decrease in AD and FTD ($|d| \approx 0.5$–0.8; consistent with posterior Alpha slowing [10, 11]); widespread Beta elevation in MDD ($|d| > 1.0$; consistent with resting-state Beta hypersynchrony [19, 18]); and temporal/frontal Alpha elevation in SCZ ($|d| \approx 0.3$; consistent with Alpha lateralization [20]). The MDD effect sizes ($|d| > 1.0$) are substantially larger than those for AD, FTD, or SCZ (0.26–0.82), which explains the stronger LPSO classification performance in MDD. Crucially, even this large effect size does not eliminate overlap inflation or lucky-fold variance.

Large inter-subject variability in epoch success rate means that the accuracy of any small held-out cohort is dominated by a few individuals, directly enabling the lucky-fold effect.

Table 14: Appendix C — Trap 3: Objective Mismatch (epoch vs. subject-accuracy gap). T-test and PCA at $P = 6$ and $P = 2$ (all models, best HP). Gap = epoch accuracy − subject accuracy; negative = subject accuracy exceeds epoch accuracy. Min/Max is range across 4 models. Note: this table reports per-model *means* across folds (range view), while main-body Tables 5–7 use *medians* (summary view); subject accuracy is quantized at the fold level so median differs from mean for that metric.

			Ep. Acc.		**Sub. Acc.**		**Gap (pp)**	
Dataset	**Pipeline**	**P**	Min	Max	Min	Max	Min	Max
AD	T-test	6	0.699	0.727	0.793	0.857	−14.1	−6.6
AD	T-test	2	0.660	0.682	0.720	0.770	−11.0	−5.4
AD	PCA	6	0.549	0.596	0.550	0.623	−5.1	+3.6
AD	PCA	2	0.533	0.560	0.490	0.600	−6.1	+4.3
FTD	T-test	6	0.605	0.631	0.623	0.650	−4.2	−0.5
FTD	T-test	2	0.604	0.624	0.580	0.620	+0.4	+4.1
FTD	PCA	6	0.542	0.555	0.500	0.520	+2.2	+4.5
FTD	PCA	2	0.556	0.558	0.500	0.500	+5.6	+5.8
MDD EC	T-test	6	0.796	0.824	0.830	0.883	−5.9	−3.4
MDD EC	T-test	2	0.792	0.823	0.800	0.870	−5.1	−0.8
MDD EC	PCA	6	0.487	0.642	0.500	0.653	−1.3	−0.3
MDD EC	PCA	2	0.484	0.666	0.500	0.710	−4.4	+0.2
MDD EO	T-test	6	0.748	0.794	0.777	0.847	−5.4	−2.8
MDD EO	T-test	2	0.768	0.867	0.760	0.940	−8.5	+0.8
MDD EO	PCA	6	0.494	0.630	0.500	0.653	−2.3	−0.1
MDD EO	PCA	2	0.500	0.700	0.500	0.740	−4.0	+0.3
SCZ	T-test	6	0.556	0.597	0.587	0.690	−10.1	−2.9
SCZ	T-test	2	0.520	0.566	0.440	0.650	−8.4	+8.0
SCZ	PCA	6	0.473	0.557	0.497	0.567	−2.7	+0.4
SCZ	PCA	2	0.464	0.537	0.500	0.500	−3.6	+3.7

Table 15: Appendix D — **Trap 1 at the subject level** (T-test pipeline, all 5 cohorts). Subject accuracy is the fraction of held-out subjects classified correctly by majority vote, pooled across all 4 models, 3 hyperparameter settings, and 50 LPSO folds ($P = 6$) for the disjoint condition (600 evaluations per disease) or 10 W_C seeds for the overlap condition (120 evaluations per disease). For W_C overlap, per-evaluation subject accuracy is computed from the saved `test_predictions.parquet` via majority vote per subject. The subject accuracy inflation pattern mirrors the epoch accuracy version (Table 5): every disease shows real subject-level inflation under overlap, ranging from +10.6 pp (MDD EC) to +32.5 pp (SCZ).

Dataset	**Sub. Acc.** $P = 6$ **(disj.)**	**Sub. Acc. overlap**	**Inflation (pp)**
AD	0.796	0.916	+12.1 pp
FTD	0.614	0.878	+26.4 pp
MDD EC	0.864	0.970	+10.6 pp
MDD EO	0.806	0.944	+13.8 pp
SCZ	0.625	0.950	+32.5 pp

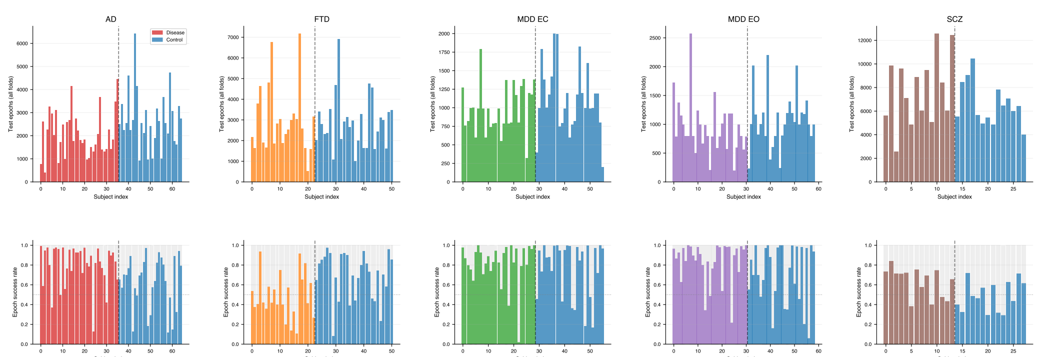

Figure 16: Per-subject test epoch counts (top) and epoch success rates (bottom) under LPSO $P = 6$ T-test, MLP. Large inter-subject heterogeneity in success rate is the mechanism behind Trap 2 — a held-out cohort of 2–6 subjects will have accuracy dominated by whichever subjects it happens to contain.

Table 16: Appendix E — **Trap 2 at the subject level** (T-test pipeline, all 5 cohorts). Each cell aggregates 50 LPSO folds. Mean subject accuracy is the simple mean across all (model, HP, fold) evaluations. IQR is computed on per-fold pooled subject accuracy (averaged across the 4 models $\times$ 3 HPs within each fold), then taken across the 50 folds. **Lucky fold = a fold where at least one (model, HP) configuration reached single-fold subject accuracy = 1.000** — meaning a single-fold study could cherry-pick that fold and report perfect subject classification. The IQR ratio mirrors the epoch accuracy pattern (Table 6) for 4 of 5 diseases; MDD EO is the exception: at the subject level, fold variability does not inflate from $P = 6$ to $P = 2$ even though it does at the epoch level. The lucky-fold rate is striking at $P = 2$: 62–94% of folds in every disease.

	Mean Sub. Acc.		**Sub. Acc. IQR (pp)**			**% lucky folds**	
Dataset	$P = 6$	$P = 2$	$P = 6$	$P = 2$	**IQR ratio**	$P = 6$	$P = 2$
AD	0.796	0.723	14.9	43.8	2.9×	56%	74%
FTD	0.614	0.590	12.5	16.7	1.3×	6%	62%
MDD EC	0.864	0.849	18.8	26.0	1.4×	70%	84%
MDD EO	0.806	0.892	16.3	15.6	1.0×	56%	94%
SCZ	0.625	0.554	19.4	25.0	1.3×	18%	62%

AI Disclosure

Large language models (Claude 4, Claude Sonnet 4.6, Google Gemini) were used as assistive tools during idea development, coding, data analysis, and manuscript drafting. All content was verified and finalized by the author, who takes full responsibility for the accuracy and integrity of this work.

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